

PROFESSIONAL SUMMERY

Robotics AI Researcher specializing in **robot learning, geometric deep learning, Robotics Foundation Models (RFMs), Vision-Language-Action (VLAs), large behavior models, and safe imitation learning**. Experienced in developing end-to-end robotic learning pipelines for **long-horizon task execution under partial observability**, with emphasis on **generalization, sim-to-real transfer, sample efficiency, and robustness under distribution shift**. Designed and deployed a modular Robotics Foundation Model integrating hierarchical decision-making, skill learning, and closed-loop execution on real robotic platforms, including **Boston Dynamics Spot and UR**. Strong background in **reinforcement learning (model-based and model-free), imitation learning (behavior cloning, diffusion policies), representation learning, and transfer learning**, with expertise in incorporating **geometric inductive biases** such as invariance and equivariance to improve sample efficiency and generalization. Proven ability to bridge research and deployment by building **scalable robotic learning** validated in both simulation and real-world environments.

EXPERIENCE

Research Assistant - The Helping Hands Lab

Boston, MA

August 2023 – Present

❖ **Bridging Semantics and Geometry - Equivariant Representations for Generalizable VLAs:**

- Proposed a **geometric adapter on VLM backbone and a dual-stream semantic-geometric token fusion** architecture for improving action prediction while preserving pretrained semantic understanding.
- Incorporating **geometric inductive biases into VLAs** through **geometry-aware representation learning and policy learning**.
- **Improving VLAs' robustness** to viewpoint changes, object pose variations, and **distribution shifts while preserving the semantic and language understanding of pretrained VLM**

❖ **Latent-Space Safety for VLAs - From Detection to Prediction:**

- Proposed a **latent-space safety method for VLAs by detecting failure-prone states** directly from the model's internal representations.
- Designed a **safety function and latent dynamic model** that can predict potential failures before they physically occur, enabling earlier **recover policy intervention** during robot execution.
- **Improving the safety and robustness of VLA-based robot policies** through failure prediction and safe recovery policies.

❖ **A Unified Learnable Embodied Robotic Agent capable of Zero-shot completion of Long-horizon tasks in Unseen environments:**

- Developed an **end-to-end cognitive robotics pipeline** integrating task interpretation, task-relevant component identification, active perception, state estimation, reasoning, planning, decision-making, closed-loop execution, and failure detection and diagnosis.
- Integrated **linguistic abstractions** to bridge high-level policies with goal-conditioned low-level execution.
- Built a **continual lifelong learning framework** leveraging a dynamically extensible skill library.
- Developed a **failure detection and diagnosis system** to identify unsuccessful skill executions and their root causes, improving overall task success by **~12%** and goal-condition success by **~25%** in long-horizon tasks.
- Proposed and developed a **Modular Robotics Foundation Model (MRFM)** integrating perceptual feedback, task history, and failure diagnosis to enable a fully learnable robotic cognition framework.
- Developed a **hierarchical POSMDP-based MRFM** separating high-level decision-making (what to do) from low-level execution (how to do).
- Achieved **~50% improvement in task success** and **~67% improvement in goal-condition success** compared to the **BUTLER**.
- Currently evaluating the framework against VLAs including **OpenVLA, $\pi 0$, and $\pi 0.5$** .

❖ **Generalizable Hierarchical Skill Learning:**

- Developed language-conditioned **high-level skills for Boston Dynamics Spot**, including:
 - ✓ **Active Search:** Environment Exploration, Prioritizing likely receptacles, Detection, State Estimation.
 - ✓ **Object Pickup:** Navigating toward the object, adjusting pose and distance, and grasping.
 - ✓ **Object Placement:** Placing a held object in/on/inside a specified receptacle

✓ **Opening/Closing:** Cabinets, Drawers, Microwave, and Fridges.

- **Integrated the Spot SDK with Multimodal Foundation Models** to ground high-level policies in real-world execution and bridge to language-conditioned low-level skills.
- Trained low-level parameterized skills (RL, IL, Diffusion policy, Equivariant policy).
- Developing high-level and low-level skills for **Unitree G1** and **Universal Robots** to enable robot-agnostic generalization.
- Extending the framework to support **online skill acquisition**, enabling robots to learn new skills dynamically when required by a task — no more offline-only skill learning.

❖ **Developed a task-driven active perception framework that:**

- **Identifies task-relevant** objects, receptacles, and environmental elements.
- **Searches for and detects** desired unseen objects in partially observable environments.
- **Estimates the states** of task-relevant components in unstructured scenes.
- **Evaluates skill termination** (success/failure).
- **Determines feasible next skills** at the current belief state (reducing the action space for the policy head).
- Demonstrated active perception improvements, leading to: **~60% increase in downstream skill selection accuracy**, **~25% improvement in task success**, and **~40% improvement in goal-condition success**.

❖ **Implemented and validated the full framework across both real robotic hardware and simulation:**

- Evaluated the framework against **AI2-THOR - ALFRED benchmark**, demonstrating a **significantly improved success rate and strong generalization capabilities**.
- **Deployed the framework on Boston Dynamics Spot**
- Deploying **the system on Unitree G1 and universal robots**.

ML Research Engineer - DeepCharge, CoinPela

Boston, MA

March 2024 - Sept 2024

- ❖ **Developed DQN-based models** for cryptocurrency price prediction using multi-scale candlestick data, improving short-horizon forecasting stability.
- ❖ Explored **model-based RL and world model approaches (MuZero, Dreamer)** for financial time-series modeling, evaluating sample efficiency and predictive robustness.
- ❖ Designed and implemented **a robust parallel OCR pipeline** utilizing a fisheye camera for wide-angle screen capture. This pipeline combined **PaddleOCR and CLIP** to accurately extract battery percentage and other critical information from parallel streams of diverse cell phone screens.
- ❖ Engineered **the embedded ESP32-Jetson Xavier NX communication pipeline** to enable synchronized sensing, data streaming, and real-time preprocessing, addressing distortion correction for large-scale deployment.
- ❖ Achieved **95% text extraction accuracy with the OCR pipeline**, validating the system by processing data from **over 1,000 cell phones in a real-world setting**.

Research Assistant - Lab for Learning and Planning in Robotics

Boston, MA

May 2022 – July 2023

- ❖ Conducted advanced research on **robotics learning, focusing on generalization, sample efficiency, and sim-to-real transfer**.
- ❖ Reformulated prior **belief-space exploration algorithms using learning-based approaches**, specifically exploring **Sim-to-Seg/Seg-to-Real**, skill learning, and language-based hierarchical decision-making.
- ❖ **Skill Learning — Skill-Based Model-Based Reinforcement Learning:**
 - **Jointly trained a skill library and skill model using VAEs on large task agnostic offline datasets**.
 - Achieved **efficient online high-level behavior learning by leveraging the learned skill model** as a world model
 - Effectively **fine-tuned representation and skill models (trained offline)** to ensure robust sim-to-real transfer.
- ❖ Extended autonomous exploration research to **end-to-end learning off-road autonomous driving (Sim-to-Real)**, specifically addressing **critical challenges in safety, transparency, and uncertainty**.
- ❖ **Accelerated MuZero/AlphaZero training by 3×** through the introduction of a novel, compact board-state representation, **reducing policy search complexity from $O(P(n))$ to $O(n)$** .

- ❖ Developed hierarchical algorithms for **autonomous mobile robot exploration under uncertainty (continuous belief spaces)**
- ❖ Extended and integrated **Feedback-based Information Road Map (FIRM)** framework in an **efficient and safe algorithm for exploration in continuous belief spaces**.
- ❖ **Proposed and Developed FIRM-MDP Hierarchical Exploration framework**, designed to manage complexity in continuous belief spaces:
 - **High-level decision making (Global Planner)**: Formulated as a **dynamic-programming TSP solver on the FIRM graph in continuous belief space**, selecting the optimal sequence of regions to explore large multi-region environments.
 - **Low-level decision making (Local Planner)**: A **continuous-belief-space exploration planner** that ensures safe and efficient local exploration within each region, accounting for **uncertainty in the exploration cost function to maintain safety**.
- ❖ Demonstrated that the hierarchical FIRM-MDP framework **significantly outperforms classical baselines (Frontier, NBV) in long-horizon exploration efficiency and robustness under uncertainty**.
- ❖ Introduced and developed a **robust region-to-region navigation algorithm ("Transfer")** for reliable and safe motion during exploration tasks, increasing navigation reliability by **~45% in cluttered** environments under motion and observation uncertainty.
- ❖ Implemented algorithms on a **non-holonomic mobile robot (Melon)** using **DFL for stabilization, LQG for path following**, OpenCV, an RGB camera with **EKF for estimation and localization**, and **LIDAR with probabilistic occupancy grid mapping**

TECHNICAL SKILLS

Programming: Python, C/C++, Java, MATLAB, Julia

AI/ML/Robotics Learning: RL, IL, Deep Learning, Geometric Deep Learning, VLAs, RFMs, Large Behavioral models, VLM Robotics, Diffusion Models/Policies, Equivariant Policies, Skill Abstraction & Learning, World Models & Skill Dynamics Learning, Generative AI, Representation Learning, Transfer Learning, Large Foundation Models, Agentic AI, Computer Vision

Foundation Models: VLAs, VLMs, LLMs, Robotics Foundation Model, Transformers, LoRA/QLoRA, RLHF/GRPO, FlashAttention, HuggingFace Transformers/Diffusers/Accelerate, Memory-based agents

Robotics: SLAM, Motion Planning, Perception, Bayesian Filters & Estimation, Hierarchical Decision-Making under uncertainty (POMDP/POSMDP), Control Theory, ROS2, Boston Dynamics Spot SDK, Unitree G1 SDK

Robots: Boston Dynamics Spot, Unitree G1 Humanoid, UR Arm Series, Mobile Robots

Simulators & Environments: AI2-THOR, Habitat, PyBullet, OpenAI Gym, Isaac Lab, MuJoCo, Gazebo, Webots, MimicGen, RoboSuite, RoboMimic, Open-Spiel, ALFRED Benchmark, LIBERO Benchmark

EDUCATION

Khoury College of Computer Sciences, Northeastern University, Boston, MA

M.S in Artificial Intelligence

- Related Courses: Advanced RL, Deep Learning, ML, AI, Foundation Models, Agentic AI, Algorithms
- Thesis: Lifelong Learning by Transferable Linguistic Abstraction

K.N. Toosi University of Technology (KNTU), Tehran, Iran

M.S in Mechatronics (Robotics)

- Related Courses: Advanced Robotics, Deep Learning, Mechatronics, Control Theory(Nonlinear & Adaptive, Optimal & Stochastic, MPC, Multi-agent Cooperative)
- Thesis: FIRM-MDP Hierarchical Exploration - Autonomous Mobile Robot Exploration under Uncertainty